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Module 9 Assignment 1: Natural Language Processing with Disaster Tweets

For the Natural Language Processing with Disaster Tweets Kaggle Competition, the Management/Research Question of interest is to develop a model that can successfully identify instances of tweets related to real disasters against other unimportant tweets and to classify them correctly. This can allow for improvement in response to disastrous events and to more quickly identify areas requiring more attention.

In using the data, I did not use the keywords as less than 5% of the data had entries for it, and the locations included a lot of information that was not necessarily related to real locations even though closer to 50% of the data had values. I started preprocessing by applying lexical analysis (cleaning the text data) and I removed hyperlinks, user handles, hashtags, ampersands, and the like from the data. Abbreviations were expanded. I also removed all punctuation marks, which may have proved detrimental to the results, as it caused the loss of sentence structure within individual tweets. All the text was then filtered so that only characters A-Z, a-z, and 0-9 were allowed to remain in the data before it was tokenized into lists. I displayed the data to a great extent so that I could find necessary modifications and to continue cleaning the text. All of the text was then converted to lowercase letters only. I used the NLTK library to eliminate stop-words (somewhat meaningless words) from the text in order to speed up training and to simplify the text into its most important/meaningful components. Afterwards, I lemmatized the tweet texts so that words that have similar meanings would be grouped together. For example, the words “great”

and “good” would be joined up together, this helps simplify and lower the total number of words in the dataset.

The total number of words as tokens was generated to reach up to 500 words, and this led to instances of tweets with up to 15 columns for the training data and 13 columns for the testing data. Any tweets not reaching the full amounts were then padded with zeros to reach 15 columns of total size so that they could be used to train the different models. After selecting the best model (model 2 with 0.77290) , I changed the number of tokens to 1000 words, and this gave me a better result against the Kaggle hidden test targets. (0.78118)

I created three Long Short-Term Memory (LSTM) models using Keras through Tensorflow. All models used only one LSTM layer. I tested higher complexity models but they were very quickly overfitting the data and the validation loss would start to increase after only 2 or 3 epoch. The 3 models had different hyperparameter numbers for the initial Embedding layer and for the LSTM layer as these were the most meaningful changes that would still provide a better model. The number of batches was tested multiple times randomly and I decided that smaller number closer to batches of 10 were the ones providing the least overfitting of the data based on the resulting information. The number of epochs was also limited as to avoid overfitting. LSTM is a very powerful method and it quickly overfits the data. Other layers were tested (such as adding extra dense layers) but the accuracy dramatically decreased to around and below 60% and therefore it was not beneficial to maintain these.

In order to further improve this algorithm, the best possible way would be to reimplement punctuation so that the impact of the order of words in the text would be more pronounced, and to implement a randomized grid search that would bring about better numbers for hyperparameter tuning. Another way to improve it would be to use a mechanism

to correct any orthographical errors in the data and to convert letters with accents into unicode instead of simply erasing them from

Appendix A: Kaggle Username and LSTM Submission Score (0.78118)

 kaggle.com/melekmizher						
692	Collin Smith		0.78148	3	2d	
693	goerman		0.78148	2	3h	
694	Patrick #6		0.78118	1	1mo	
695	Melek Mizher		0.78118	5	1s	
 Your Best Entry! Your most recent submission scored 0.78118, which is an improvement of your previous score of 0.77290. Great job! Tweet this						
696	Julie S		0.78087	1	2mo	
697	Red John		0.78087	1	1mo	
698	advait bakshi		0.78087	1	1mo	

Appendix B: Python Notebook and Data Visualization/Data Tables

```
import pandas as pd
import numpy as np
from numpy import std, absolute, mean
import seaborn as sns
import sklearn as sk
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
import datetime
import tensorflow as tf
from tensorflow.python.client import device_lib #GPU Check
import tensorflow.keras #keras
from tensorflow.keras import layers
```

```

from tensorflow.keras import Sequential, Input, Model
from tensorflow.keras.layers import Dense, Dropout, Flatten, Input, Add,
Activation, ZeroPadding2D
from tensorflow.keras.layers import LSTM, BatchNormalization, Conv2D,
AveragePooling2D, MaxPooling2D, GlobalMaxPooling2D
from tensorflow.keras.callbacks import EarlyStopping, ReduceLROnPlateau,
ModelCheckpoint #use for early stopping
from tensorflow.keras.initializers import glorot_uniform, he_uniform #to
initialize random weights for filters
from tensorflow.keras.preprocessing.image import ImageDataGenerator
from tensorflow.keras.preprocessing import image as image_utils
from tensorflow.keras.applications.imagenet_utils import preprocess_input,
decode_predictions
from tensorflow.keras.models import Sequential, Model, load_model
from tensorflow.keras import utils
from tensorflow.keras.utils import get_file, plot_model, to_categorical,
model_to_dot

```

In [2]:

```

from IPython.core.interactiveshell import InteractiveShell
InteractiveShell.ast_node_interactivity = "all"

```

In [3]:

```
%pwd
```

Out[3]:

```
'/Users/melekmizher/Desktop/NLP with Disaster Tweets'
```

In [4]:

```

train = pd.read_csv("train.csv")
test = pd.read_csv("test.csv")

```

In [5]:

```

train.head()
test.head()

```

Out[5]:

	i d	keywor d	locatio n	text	targe t
0	1	NaN	NaN	Our Deeds are the Reason of this #earthquake M...	1
1	4	NaN	NaN	Forest fire near La Ronge Sask. Canada	1
2	5	NaN	NaN	All residents asked to 'shelter in place' are ...	1
3	6	NaN	NaN	13,000 people receive #wildfires evacuation or...	1
4	7	NaN	NaN	Just got sent this photo from Ruby #Alaska as ...	1

Out[5]:

	id	keyword	location	text
0	0	NaN	NaN	Just happened a terrible car crash
1	2	NaN	NaN	Heard about #earthquake is different cities, s...
2	3	NaN	NaN	there is a forest fire at spot pond, geese are...
3	9	NaN	NaN	Apocalypse lighting. #Spokane #wildfires
4	11	NaN	NaN	Typhoon Soudelor kills 28 in China and Taiwan

In [6]:

train

Out[6]:

	id	keyword	location	text	target
0	1	NaN	NaN	Our Deeds are the Reason of this #earthquake M...	1
1	4	NaN	NaN	Forest fire near La Ronge Sask. Canada	1
2	5	NaN	NaN	All residents asked to 'shelter in place' are ...	1
3	6	NaN	NaN	13,000 people receive #wildfires evacuation or...	1
4	7	NaN	NaN	Just got sent this photo from Ruby #Alaska as ...	1
...	...	...	...	...	...
7608	10869	NaN	NaN	Two giant cranes holding a bridge collapse int...	1
7609	10870	NaN	NaN	@aria_ahrery @TheTawniest The out of control w...	1
7610	10871	NaN	NaN	M1.94 [01:04 UTC]?5km S of Volcano Hawaii. htt...	1

761	1087	NaN	NaN	Police investigating after an e-bike collided ...	1
1	2				

761	1087	NaN	NaN	The Latest: More Homes Razed by Northern Calif...	1
2	3				

7613 rows × 5 columns

In [7]:

```
train.keyword.value_counts()
```

Out[7]:

```
fatalities      45
deluge          42
armageddon      42
sinking         41
damage          41
..
forest%20fire   19
epicentre       12
threat          11
inundation       10
radiation%20emergency  9
Name: keyword, Length: 221, dtype: int64
```

In [8]:

```
train.location.value_counts()
```

Out[8]:

```
USA          104
New York      71
United States  50
London        45
Canada        29
...
Montréal, Québec  1
Montreal        1
İT: 6.4682,3.18287  1
Live4Heed??     1
Lincoln         1
Name: location, Length: 3341, dtype: int64
```

Lexical Analysis

In [9]:

```
import re
```

In [10]:

```
def cleanText(text):
    text = re.sub(r'@[A-Za-z0-9]+', '', text)
    text = re.sub(r'#', '', text)
    text = re.sub(r'&', ' and ', text)
    text = re.sub(r'https?:\\/\S+', '', text)
    text = re.sub(r'RT[\s]+', '', text)
    text = re.sub(r"can't", 'can not', text)
    text = re.sub(r"n't", ' not', text)
    text = re.sub(r"\s\s?|\s", ' ', text)
    text = re.sub(r"s", ' is', text)
    text = re.sub(r"re", ' are', text)
    text = re.sub(r"w/", 'with ', text)
    text = re.sub(r"ve", ' have', text)
    text = re.sub(r"m", ' am', text)
    text = re.sub(r"ll", ' will', text)
    text = re.sub(r"-", '', text)
    text = re.sub(r"!", '', text)
    text = re.sub(r",", '', text)
    text = re.sub(r";", '', text)
    text = re.sub(r":", '', text)
    text = re.sub(r"_", '', text)
    text = re.sub(r"\|", '', text)
    text = re.sub(r"@", '', text)
    text = re.sub(r"/\n", '', text)
    text = re.sub(r"\?", '', text)
    text = re.sub(r"^[A-Za-z0-9]", " ", text)
    return text
```

In [11]:

```
train['text'] = train['text'].apply(cleanText)
train['text'] = train['text'].str.lower()
```

In [12]:

```
test['text'] = test['text'].apply(cleanText)
test['text'] = test['text'].str.lower()
```

In [13]:

test

Out[13]:

	id	keyword d	location	text
0	0	NaN	NaN	just happened a terrible car crash
1	2	NaN	NaN	heard about earthquake is different cities sta...
2	3	NaN	NaN	there is a forest fire at spot pond geese are ...

3	9	NaN	NaN	apocalypse lighting spokane wildfires
4	11	NaN	NaN	typhoon soudelor kills 28 in china and taiwan
...	...	...	...	...
325	1086	NaN	NaN	earthquake safety los angeles safety faste...
8	1			
325	1086	NaN	NaN	storm in ri worse than last hurricane my city...
9	5			
326	1086	NaN	NaN	green line derailment in chicago
0	8			
326	1087	NaN	NaN	meg issues hazardous weather outlook hwo
1	4			
326	1087	NaN	NaN	cityofcalgary has activated its municipal emer...
2	5			

3263 rows × 4 columns

In [14]:

```
pd.options.display.max_rows = 500
```

In [15]:

```
#pd.describe_option('display')
```

In [16]:

```
pd.describe_option('display.max_colwidth')
pd.set_option('display.max_colwidth', 110)
display.max_colwidth : int or None
```

The maximum width in characters of a column in the repr of a pandas data structure. When the column overflows, a "..." placeholder is embedded in the output. A 'None' value means unlimited.
[default: 50] [currently: 50]

In [17]:

```
train.text.head(500)
```

Out[17]:

```
0          our deeds are the reason of this earthquake may allah forgive
us all
1          forest fire near la ronge sask canada
2  all residents asked to shelter in place are being notified by officers no other
evacuation or shelter in ...
```


3 13000 people receive wildfires evacuation orders in
california

4 just got sent this photo from ruby alaska as smoke from wildfires pours
into a school

5 rockyfire update california hwy 20 closed in both directions due to lake county
fire cafire wildfires

6 flood disaster heavy rain causes flash flooding of streets in manitou
colorado springs areas

7 i am on top of the hill and i can see a fire in the woods

8 there is an emergency evacuation happening now in the building
across the street

9 i am afraid that the tornado is coming to our area

10 three people died from the heat wave so far

11 haha south tampa is getting flooded hah wait a second i live in south tampa what am
i gonna do what am i g...

12 raining flooding florida tampabay tampa 18 or 19 days i have
lost count

13 flood in bago myanmar we arrived bago

14 damage to school bus on 80 in multi car crash
breaking

15 what is up man

16 i love fruits

17 summer is lovely

18 my car is so fast

19 what a goooooooooaaaaaal

20 this is ridiculous

21 london is cool

22 love skiing

23 what a wonderful day

24 loooooool

25 no way i can not eat that shit

26 was in nyc last week

27 love my girlfriend

28 coooool

29 do you like pasta

30 the end

31 wholesale markets ablaze

32 we always try to bring the heavy metal

33 africanbaze breaking newsnigeria flag set ablaze in
aba

34 crying out for more set me ablaze

35 on plus side look at the sky last night it was ablaze

36 muftc they have built so much hype around new acquisitions but i doubt they will set
the epl ablaze this s...

37 inec office in abia set ablaze

38 barbados bridgetown jamaica two cars set ablaze santa cruz head of the st
elizabeth police superin...

39 ablaze for you lord d

40 check these out nsfw

41 on the outside you are ablaze and alive but you are dead
inside

42 had an awesome time visiting the cfc head office the ancop site and ablaze thanks
to tita vida for taking...

43 soooo pumped for ablaze

44 i wanted to set chicago ablaze with my preaching but not
my hotel

45 i gained 3 followers in the last week you know your stats and
grow with

46 how the west was burned thousands of wildfires ablaze in
california alone

47 building the perfect tracklist to life leave the streets
ablaze

48 check these out nsfw

49 first night with retainers in it is quite weird better get used to it i have to wear them
every single n...

50 deputies man shot before brighton home set
ablaze

51 man wife get six years jail for setting ablaze niece

52 santa cruz head of the st elizabeth police superintendent lanford
salmon has r

53 police arsonist deliberately set black church in north carolina
ablaze

54 noches elbestia sanchez happy to see my teammates and training hard
goodnight gunners

55 kurds trampling on turkmen flag later set it ablaze while others vandalized offices of
turkmen front in di...

56 truck ablaze r21 voortrekker ave outside or tambo intl cargo
section

57 set our hearts ablaze and every city was a gift and every skyline was like a kiss
upon the lips

58 they sky was ablaze tonight in los angeles i am expecting ig and fb to be filled with
sunset shots if i k...

59 how the west was burned thousands of wildfires ablaze in california alone
climate energy

60 revel in yours wmv videos by means of mac farewell ablaze wmv en
route to dvd gtxrwm

61 progressive greetings in about a month students would have set their pens ablaze
in the torch publication...

62 rene ablaze and jacinta secret 2k13 fallen skies edit mar 30
2013

63 steve these fires out here are something else california is a tinderbox and this
clown was setting my ho...

64 nowplaying rene ablaze and ian buff magnitude
edm

65 huge fire at wholesale markets ablaze

66 what time does your talk go until i do not know if i can make it due
to work

67 i can not have kids cuz i got in a bicycle accident and split my testicles it is
impossible for me to ...

68 accident on i24 w nashvilletraffic traffic moving 8m slower than
usual

69 accident center lane blocked in santaclara on us101 nb before great america
pkwy bayarea traffic

70 had a personalinjury accident this summer read our advice and see how a
solicitor can help otleyhour

71 stlouis caraccidentlawyer speeding among top causes of teen accidents
car accident tee

72 reported motor vehicle accident in curry on herman rd near stephenson involving an
overturned vehicle ple...

73 bigrigradio live accident awareness

74 i77 mile marker 31 south mooreville iredell vehicle accident ramp
closed at 8 6 118 pm

75 sleeping pills double your risk of a car accident

76 by accident they knew what was gon happen

77 traffic accident n cabrillo hwy magellan av mir 08 06 15
110358

78 i77 mile marker 31 to 40 south moorerville iredell vehicle accident
congestion at 8 6 118 pm

79 the pastor was not in the scene of the accident who was the owner of
the range rover

80 mom we did not get home as fast as we wished me why is that mom there was an
accident and some truck spil...

81 i was in a horrible car accident this past sunday i am finally able to get
around thank you god

82 can wait to see how pissed donnie is when i tell him i was in
another accident

83 truckcrash overturns on fortworth interstate click here if you have been
in a crash gt

84 accident in ashville on us 23 sb before sr 752 traffic

85 carolina accident motorcyclist dies in i540 crash with car that crossed median a
motorcycle rider travelin...

86 fyi cadfyi accident property damagenhs999 piner rd
horndale dr

87 naayf first accident in years turning onto chandanee magu from near mma taxi
rammed into me while i was ...

88 accident left lane blocked in manchester on rt 293 nb before eddy rd stop and go
traffic back to nh3a dela...

89 accident property damage piner rd horndale dr

90 it was an accident

91 fyi cadfyi accident property damagewpd1600 s 17th
st

92 8 6 201509 pm traffic accident no injury at 2781 willis
foreman rd

93 aashiqui actress anu aggarwal on her nearfatal
accident

94 suffield alberta accident

95 9 mile backup on i77 south accident blocking the right 2 lanes at exit 31 langtree rd
consider nc 115 ...

96 has an accident changed your life we will help you determine options that can
financially support life car...

97 breaking there was a deadly motorcycle car accident that happened to hagerstown
today i will have more de...

98 were you marinading it or was it an accident
99 only had a car for not even a week and got in a fucking car accident mfs can
not fucking drive
100 bahrain police had previously died in a road accident they were not
killed by explosion
101 i still have not heard church leaders of kenya coming forward to comment on the
accident issue and discipl...
102 delo scuf ps live and the game cya
103 the man who can drive himself further once the effort gets painful is the man who
will win roger bannister
104 320 ir icemoon aftershock dubstep trapmusic dnb edm
dance ices
105 there is no victory at bargain basement prices dwight david
eisenhower
106 320 ir icemoon aftershock dubstep trapmusic dnb edm
dance ices
107 nobody remembers who came in second charles
schulz
108 delo im speaking from someone that is using a scuf on xb1 most of them people will
end up getting on for p...
109 the harder the conflict the more glorious the triumph
thomas paine
110 growingupspoiled going clay pigeon shooting and crying because
of the aftershock
111 so i guess no one actually wants any free aftershock tc
112 aftershock was the most terrifying best roller coaster i have ever been on
disclaimer i have been on ve...
113 aftershock
114 320 ir icemoon aftershock dubstep trapmusic dnb edm
dance ices
115 320 ir icemoon aftershock dubstep trapmusic dnb edm
dance ices
116 320 ir icemoon aftershock dubstep trapmusic dnb edm
dance ices
117 i am seeing them and issues at aftershock
118 320 ir icemoon aftershock dubstep trapmusic dnb edm
dance ices

119 320 ir icemoon aftershock dubstep trapmusic dnb edm
dance ices

120 wisdomwed bonus 5 minute daily habits that could really improve your life how
many do you already do lif...

121 aftershock protect yourself and profit in the next global financial meltdown by
david wiedemer http

122 that moment when you get on a scary roller coaster and the guy behind you is just
screaming bloody murder ...

123 aftershock 2010 full streaming youtube

124 gt gt 15 aftershock protect yourself and profit in the next global
financial book

125 sometimes you face difficulties not because you are doing something wrong but
because you are doing someth...

126 the only thing that stands between you and your dream is the will to try and the
belief that it is actual...

127 praise god that we have ministry that tells it like it is now wdyouth
biblestudy

128 remembering that you are going to die is the best way i know to avoid the trap of
thinking you have somet...

129 tried orange aftershock today my life will never be the
same

130 i love you bb

131 aftershock

132 aftershock back to school kick off was great i want to thank everyone for making it
possible what a grea...

133 people who say it cannot be done should not interrupt those who are doing it
george bernard shaw

134 the first man gets the oyster the second man gets the shell
andrew carnegie

135 anyone need a p u tonight i play hybrid slayer ps4 eu hmu
kotc org

136 experts in france begin examining airplane debris found on reunion island french air
accident experts o ...

137 strict liability in the context of an airplane accident pilot error is a common
component of most aviation...

138 your lifetime odds of dying from an airplane accident are 1
in 8015

139 experts in france begin examining airplane debris found on reunion island french air accident experts on w...

140 awww they are on an airplane accident and they are gonna die what a cuties good job

141 family members of osama bin laden have died in an airplane accident how ironic mhmhm gov shit i suspect

142 man goes into airplane engine accident via

143 horrible accident man died in wings of airplane

29072015

144 a cessna airplane accident in ocampo coahuila mexico on july 29 2015 killed four men including a state of ...

145 horrible accident man died in wings airplane 29072015

watchthevideo

146 experts in france begin examining airplane debris found on reunion island french air accident experts on ...

147 experts in france begin examining airplane debris found on reunion island french air accident experts on w...

148 kca votejkt48id mbataweel rip binladen family members who killed in an airplane is accident

149 i almost sent my coworker nudes on accident thank god for airplane mode

150 that or they might be killed in an airplane accident in the night a car wreck politics at it is best

151 experts in france begin examining airplane debris found on reunion island french air accident experts on ...

152 this is unbelievably insane man airport airplane aircraft aeroplane runway accident freaky

153 horrible accident man died in wings of airplane

29072015

154 horrible accident man died in wings of airplane

29072015

155 usama bin ladins family dead in airplane crash naturally no accident

156 pilot dies in plane crash at car festival via crash aircraft airplane pilot death accident carfest

157 strict liability in the context of an airplane accident

158 dtn brazil experts in france begin examining airplane debris found on reunion island french air accident e...

159 experts in france begin examining airplane debris found on reunion island french air accident experts on w...

160 horrible accident man died in wings of airplane 29072015 wtf you can't believe your eyes

161 nicole fletcher one of a victim of crashed airplane few times ago the accident left a little bit trauma...

162 omg horrible accident man died in wings of airplane

163 omg i do not believe this rip bro airplane accident jet engine turbojet boeing 737

164 experts in france begin examining airplane debris found on reunion island french air accident experts on w...

165 i had a airplane accident

166 my phone looks like it was in a car ship airplane accident terrible

167 statistically i am at more of risk of getting killed by a cop than i am of dying in an airplane accident

168 airplane crashes on house in colombia 12 people die in accident

169 the shooting or the airplane accident

170 could a drone cause an airplane accident pilots worried about use of drones esp in close vicinity of airports...

171 early wake up call from my sister begging me to come over and ride with her in the ambulance to the hospital...

172 twelve feared killed in pakistani air ambulance helicopter crash

173 two air ambulances on scene of serious crash between two cars and lorry in london

174 twelve feared killed in pakistani air ambulance helicopter crash reuters yugvani

175 leading emergency services boss welcomes new ambulance charity

176 anyone travelling aberystwyth shrewsbury right now there is been an incident services at a halt just outside...

177 twelve feared killed in pakistani air ambulance helicopter crash

178 ambulance sprinter automatic frontline vehicle choice of 14 lez compliant ebay

179 new nanotech device will be able to target and destroy
blood clots

180 29 it was hella crazy 3 fights an ambulance and a couple
mosh pits

181 if i get run over by an ambulance am i lucky just saying
random thought

182 news twelve feared killed in pakistani air ambulance helicopter
crash tilnow dna

183 twelve feared killed in pakistani air ambulance helicopter
crash

184 waiting for an ambulance

185 you ok need a ambulance hahahah that was
good

186 ambulance sprinter automatic frontline vehicle choice of 14 lez
compliant ebay

187 pakistan air ambulance helicopter crash kills nine

188 nissan are you ok do you need medical assistance i can call an
ambulance if you need me to

189 ems1 ny emts petition for 17 per hour minimum wage ems
paramedics ambulance

190 twelve feared killed in pakistani air ambulance helicopter
crash

191 twelve feared killed in pakistani air ambulance helicopter
crash

192 ambulance sprinter automatic frontline vehicle choice of 14 lez
compliant ebay

193 ambulance sprinter automatic frontline vehicle choice of 14 lez
compliant ebay

194 karyn check out what is in my parking lot he said that until last year it was an
ambulance in st johns

195 when you do not know which way an ambulance is coming
from It It

196 reuters twelve feared killed in pakistani air ambulance
helicopter crash

197 twelve feared killed in pakistani air ambulance helicopter
crash

198 why is there an ambulance right outside my
work

199 this dog thinks he is an ambulance
200 happening now hatzolah ems ambulance responding with
dual sirens and
201 twelve feared killed in pakistani air ambulance helicopter
crash
202 twelve feared killed in pakistani air ambulance helicopter
crash
203 twelve feared killed in pakistani air ambulance helicopter crash
worldnews
204 twelve feared killed in pakistani air ambulance helicopter crash
worldnews
205 what is the police or ambulance number in lesotho any
body know
206 org i am surprised we still cannot standardised the clinical practice across the 11
nhs ambulance trust
207 twelve feared killed in pakistani air ambulance helicopter
crash
208 people who try to jwalk while an ambulance is passing i
hate you
209 the episode where trunks annihilated freiza is the cleanest shit ever he showed
that nigga no mercy
210 they shall be annihilated and all of their petebests dessicated and laid bare then
you shall kneel before...
211 uribe just annihilated that baseball mets
212 heysundowns were annihilated in their previous meeting with celtic indeed
its an improvement
213 mizzou has annihilated florida the past 2 seasons even ended muschamp is career
just can not compete wit...
214 annihilated abs
215 be annihilated for status education mba on behalf of a on easy street
careen eovm
216 to luka they should all die all of them everything annihilated
alois trancy
217 have a good un fella sorry i wo not be there to get annihilated
with you
218 cop pulls drunk driver to safety seconds before his car is hit by
train via
219 you must be annihilated

220 cop pulls drunk driver to safety seconds before his car is hit by
train via

221 boom your country was just entirely annihilated by a h
britain

222 you would have been annihilated so you might as well
thank

223 one thing for sure god has promised israel will not be annihilated but the horror
of iran with nukes

224 i guess that is ok for armenians since we have spent most of our history
getting annihilated

225 70 years since we annihilated 100000 people instantly and became aware that we
have the ability to annihil...

226 day 1 of tryouts went good minus the fact i stopped quickly to get a short ball and
annihilated my toenail...

227 during the 1960s the oryx a symbol of the arabian peninsula were
annihilated by hunters

228 to luka they should all die all of them everything annihilated alois
trancy

229 ready to get annihilated for the bucs game

230 people annihilated by last nights weather really philip thought you would
have forecast that

231 domain other sophistication be annihilated closely
uptotheminute feat zrnf

232 lord i seen judas priest in 2005 when rob came back scorpions as support fucking
annihilated the place ...

233 officially skipping out on fantasticfour fant4stic whatever the hashtag is it is getting
annihilated in r...

234 thus explaining why you were all annihilated but the few or in this case you the
only survivor evolved a...

235 just completely annihilated cech with paul keegan what a time
to be alive

236 who then were annihilated by the legion itself the survivors of the imperfect hybrid
project quickly for...

237 exactly that is why the lesnar cena match from summerslam last year was so great
because brock annihilat...

238 cop pulls drunk driver to safety seconds before his car is hit by
train via

239 annihilated in damascus syrian army grinds alloosh and his gang into
the manure pile

240 ok i was not completely forthright i may have also been in a food coma bc of the
kebab tahini pickles i a...

241 a fun filled happyhour at simmons bar in camden with this handsome one i got
annihilated apart from this...

242 juanny beisbol sr annihilated that ball lgm

243 hell is just a fraction of his belief of total annihilation destruction of
usa jewels

244 maybe we should call israel and tell them we are sorry are pres has sold them down
the river to annihilat...

245 evildead annihilation of civilization

246 u s national park services tonto national forest stop the annihilation of the salt river
wild horse via

247 annihilating quarterstaff of annihilation

248 world annihilation vs self transformation aliens attack to
exterminate humans

249 starmade stardate 3 planetary annihilation via

250 u s national park services tonto national forest stop the annihilation of the salt river
wild horse via

251 u s national park services tonto national forest stop the annihilation of the salt river
wild horse via

252 can you please sign and share this petition to save wild horses in
arizona

253 u s national park services tonto national forest stop the annihilation of the salt river
wild horse via

254 please check into salt river horses help stop the annihilation about to happen
without 54000 more signatu...

255 are souls punished with annihilation

256 not to mention a major contributor to the annihilation of
israel

257 we need help horses will dieplease and sign petitiontake a stand and be a
voice for them gilbert23

258 i reject the laws of the misguided false prophets imprison nations fueling
self annihilation

259 u s national park services tonto national forest stop the annihilation of the salt river
wild horse via

260 the annihilation of jeb christie and kasich is less than 24 hours away please god
allow me at least on...

261 twins we need helphorses will die please and sign petition take a stand and be a
voice for them gilbert23

262 we need help horses will die please and sign petition take a stand and be a
voice for them gilbert23

263 hey az sign this petition to save the wildhorses tantonationalforest a
singalong is in order

264 stop the annihilation of the salt river wild horses via

265 and he is the current nova in the books last i checked he was tied into the books in
2011 after rider die...

266 u s national park services tonto national forest stop the annihilation of the salt river
wild horse via

267 please sign and to save saltriverwildhorses

268 thanks dante do join us by following the zone
johnny

269 world annihilation vs self transformation aliens attack to
exterminate humans

270 u s national park services tonto national forest stop the annihilation of the salt river
wild horse via

271 u s national park services tonto national forest stop the annihilation of the salt river
wild horse via

272 i am gonna fight taylor as soon as i get there

273 ohh no fukurodani didn t survive the apocalypse bokuto feels horrible my
poor boy my ppor child

274 will there be another jocelyn birthday apocalypse

275 rt janenelson097 stephenscifi adaptation watch charlie human is apocalypse now
now optioned for film scien...

276 apocalypse please

277 another hour it is august 05 2015 at 0802pm here is red rover zombie apocalypse
2014 internetradio colleg...

278 i feel like me doing a pullup is one of the stages of the
apocalypse

279 she is kinda hot played on the radio today what is next disease to all the
apocalypse has started everyon...

280 but if it is the apocalypse lol gf m8

281 i know it is a question of interpretation but this is a sign of the apocalypse
i called it

282 julie r is the apocalypse version of romeo juliet
warmbodies

283 the apocalypse is upon us

284 rt fittscott minecraft night lucky block mod bob apocalypse wither 2 0 and more
mod showcase popularmm...

285 and so it begins day one of the snow
apocalypse

286 rt ourmothermary short reading apocalypse 211023 in the spirit the angel took me
to the top of an enorm...

287 candylit imagine sarumi in a zombie apocalypse fighting back to back heart to heart
conversations over the...

288 rt zonewolf123 i liked a youtube video minecraft night lucky block mod bob
apocalypse wither 2 0 and m...

289 and that is because on my planet it is the lone audience of the
apocalypse

290 dad bought a dvd that looks like a science doc on the front but i read the back and it
is actually about t...

291 it is indeed if the apocalypse comes this week i know where i
will be

292 go look at grizzly peak right now it looks like the beginning of an dystopian
apocalypse movie

293 my niece just asked me would you be scared if there was an
apocalypse here

294 there is a storm over cairo in the latest xmen apocalypse set
photo via

295 minecraft night lucky block mod bob apocalypse wither 2 0 and more mod
showcase popularmmos via

296 shot through the heart xv you are going to totally give love a bad name with
this heart pierc

297 rt geekapocalypse 4pm gmt hesse plays dark souls 2 day
9 etcpb

298 i know where to go when the zombies take over

299 the latest from reveals storm is a queen in apocalypse

300 shadow boxing the apocalypse

301 short reading apocalypse 211023 in the spirit the angel took me to the top of an
enormous high mountain...

302 enjoyed liveaction attack on titan but every time i see posters i am reminded how
freshly clean and coiffe...

303 i liked a video minecraft night lucky block mod bob apocalypse wither 2 0
and more mod showcase

304 pbban temporary300 avysss armageddon do not kill flags only
fast xp for reason

305 pbban temporary300 russaky89 armageddon do not kill flags only
fast xp for reason

306 official vid doublecups gt gt gt gt prod gt gt
armageddon

307 ouvindo peace love and armageddon

308 best movie you have ever seen armageddon

309 bed time do not wake me up unless revolution or
armageddon start

310 red faction armageddon microsoft xbox 360 2011 full read
by ebay

311 you already know how this shit goes world series or
armageddon

312 love truelove romance lith voodoo seduction astrology rtrlotz 911 apocalypse
armageddon 1008pla

313 im gonna beat armageddon as hsu hao just got a flawless on
my first try

314 ben affleck i know there is a wife kids and other girls but i can not help it i have
loved him sinc...

315 if i d have had a long coat to hand i d have worn it the certainty of armageddon
bears a sense of occasi...

316 your phone is spying on you hidden back door nsa data mining software the
financial armageddon blog

317 pbban temporary300 hyiderghost2 armageddon do not kill flags only
fast xp for reason

318 love truelove romance lith voodoo seduction astrology rtrlotz 911 apocalypse
armageddon 1008planet

319 pbban temporary300 fighterdena armageddon do not kill flags only
fast xp for reason

320 photo sketch i did based on the a taste of armageddon episode of
startrek tos

321 armageddon

322 good luck to the tomorrow night get some coefficient
points plz

323 armageddon

324 you are another one for the history books thank the justice department and by the way i have not paid i...

325 vladimir putin issues major warning but is it too late to escape armageddon

326 god is kingdom heavenly gov t will rule over all people on the earth after armageddon

327 i b entertainment lot of 8 brucewillis movies dvd die hard 1 2 12 monkeys armageddon sixth ebay auction

328 let s talk some more about your goof guild saunders come right up here on stage

329 my niece is gaining the ability to stand i am getting prepared for toddler apocalypse armageddon

330 check out preppers doomsday must have library collection on cd shtf preppertalk survival 2a prepper via

331 again eep thought of you yesterday when i saw that hella scary hail armageddon

332 so the ahamedis think the messiah had already come 125 years ago where is armageddon where is the dajaal w...

333 sadly how windows 10 reveals microsoft is ethics armageddon

334 armageddon averted by el patron ultimalucha

335 if you have the time 23 hours the latest series about ww1 blueprint for armageddon is extremely impres...

336 european fitba till christmas armageddon

337 christians united for israel cufi jews should convert soon or die by armageddon us

338 official vid gt doublecups gt gt gt gt prod gt gt armageddon

339 tomorrow is the day we start armageddon
preseasonworkouts

340 lee does comedy working class tories prepare for your armageddon
interestraterise

341 9 charts prove financial crisis part 2 has begun the financial armageddon economic collapse blog tracks tr...

342 paul craig roberts vladimir putin issues major warning but is it too late to escape a brics roberts r...

343 you are on stage now right under the lights is not it funny where do you get the goofballs with which you...

344 official vid thereal gt gt gt gt gt gt gt gt gt armageddon comin
soon

345 well done celtic fingers crossed for aberdeen tomorrow night
armageddon eh

346 beyonce is my pick for fan army beyhive 77

347 one direction is my pick for fan army directioners

x1402

348 5 seconds of summer is my pick for fan army

5sosfam

349 22 beyonce is my pick for fan army beyhive

350 17 beyonce is my pick for fan army beyhive

351 one direction is my pick for fan army directioners

x1411

352 seeing that army of whitewalkers was the very first thing that has slightly
intrigued me on got so far

353 build your own kingdom and lead your army to victory start g this
friend code lzktjnox

354 salvation army hosts rally to reconnect fathers with children the salvation army is
hosting a back to scho...

355 vote for directioners vs queens in the 5th round of the
fanarmyfaceoff

356 but if you build an army of 100 dogs and their leader is a lion all dogs will
fight like a lion

357 show me a hero tv review

358 one direction is my pick for fan army directioners

x1392

359 infantry mens lume dial army analog quartz wrist watch sport blue nylon
fabric full rea

360 one direction is my pick for fan army directioners

x1441

361 victorinox swiss army date women s rubber mop watch

241487

362 drayesha4 indiakomuntorjawabdo indian army ki

363 5 seconds of summer is my pick for fan army

5sosfam 1

364 beyonce is my pick for fan army beyhive 78

365 one direction is my pick for fan army directioners

x1386

366 you da one mtvsummerstar videoveranomtv mtvhottest britney
spears lana del rey

367 one direction is my pick for fan army directioners

368 stony jackson is america is last hope as he leads an army of felons thus and army
rejects against the army...

369 wwi wwii japanese army navy military japan leather watch war mido ww1 2
full read by ebay

370 beyonce is my pick for fan army beyhive 72

371 7 beyonce is my pick for fan army beyhive

372 beyonce is my pick for fan army beyhive 66

373 6 beyonce is my pick for fan army beyhive

374 potus appoints brig gen richard g kaiser as member of the mississippi river
commission learn more abou...

375 what a violent country get the army involved to help control the killings and bring
back peace to the poo...

376 wwi wwii japanese army navy military japan leather watch war mido ww1 2
full read by ebay

377 wwi wwii japanese army navy military japan leather watch war mido ww1 2
full read by ebay

378 one direction is my pick for fan army directioners

x1434

379 one direction is my pick for fan army directioners

380 two jewish terrorists charged in historicchurch arson the
ugly truth

381 spokane authorities say they are struggling to solve arson cases like today
is on hamilton

382 thousands attend a rally organized by peace now protesting the arson attack
that took the life of an

383 add familia to the arson squad

384 another fake hate crime lesbians burn their own house down
what else is new

385 los angeles times arson suspect linked to 30 fires caught in northern
newsintweets

386 mourning notices for stabbing arson victims stir politics of grief
in israel

387 mourning notices for stabbing arson victims stir politics of grief
in israel

388 the sound of arson

389 owner of chicagoarea gay bar admits to arson scheme
lgbt
390 wait what
391 arson suspect linked to 30 fires caught in northern
california
392 trial date set for man charged with arson burglary
393 after death of palestinian toddler in arson attack israel cracks
down on jewish
394 palestinian teen killed amid protests against arson
attack
395 kisii police in kisii hunt for students over failed arson plot police in kisii hunt for
students count...
396 mariah getting thick in the shoulders poor girl
397 mourning notices for stabbing arson victims stir politics of grief in israel posters
for shira banki...
398 relaxinpr miprv latimes arson suspect linked to 30 fires caught in
northern california
399 jewish leaders prayed at the hospital where a palestinian family is being
treated after arson via
400 owner of chicagoarea gay bar admits to arson scheme
via lgbt
401 ae angel or arson
402 kisii police in kisii hunt for students over failed arson plot police in kisii hunt for
students count...
403 mourning notices for stabbing arson victims stir politics of grief
in israel
404 owner of chicagoarea gay bar admits to arson scheme
via lgbt
405 owner of chicagoarea gay bar admits to arson scheme
lgbt
406 mourning notices for stabbing arson victims stir politics of grief
in israel
407 owner of chicagoarea gay bar admits to arson
scheme lgbt
408 arson suspect linked to 30 fires caught in northern california
via
409 tennessee lesbian couple faked hate crime and destroyed own home
with arson lesbian

410 arson suspect linked to 30 fires caught in northern california los
angeles times

411 arson suspect linked to 30 fires caught in northern
california

412 301dc goldrush i hate white people mo
413 nowplaying arsonist mc so impressed
414 alleged east bay serial arsonist arrested

415 just a minor citation for possession of a decriminalized substance im not
facing any time

416 suspected serial arsonist arrested in calif
417 arson suspect linked to 30 fires caught in northern
california

418 arsonist the legal system never forgets
419 rmg u on dick

420 bloor ossington arsonist also burned a mattress on
northumberland st cbcto

421 where is my arsonist at
422 if you do not have anything nice to say you can come sit
with me

423 vegetarian vegan video shows arsonist torching popular bk restaurant strictly
vegetarian govegan unite...

424 arsonist arrested for setting many fires watch tonight s other headlines
nightbeat 2minutemix

425 video captures man removing american flag from long beach ca home
burning it arsonist sought

426 my trick is to think about nasty things
427 spotlight take me to paradise by arsonist mc
wniagospel

428 who makes these
429 on town of salem i just melted ice cube bc im the
arsonist d

430 arsonists being blamed for a blaze at a plastics recycling business in adelaide
reports 7newsadl

431 i be on that hotboy shit
432 zodiac girl feat trey dupree produced by sparkz beatz chuck da
arsonist

433 arsonist lmfaol
434 alleged east bay serial arsonist arrested sanfrancisco

435 doofus im jokin still cant be on moves
436 arsonist i guess u can say that it is just some shit i was
thinking about
437 arsonist sets nyc vegetarian restaurant on fire police
newyork
438 i liked a video from town of salem how to win as the
arsonist
439 rmg whats cracking cuz
440 smoke good fuck eat drink drive nice car wear all green
mink
441 kill i got court the day after earl
442 arsonist lmao but real live you should go
443 owner of chicagoarea gay bar admits to arson scheme frank elliott pleaded guilty to
hiring an arsonist to ...
444 trusting iran to stop terrorism is like inviting an arsonist to join the fire
brigade telegraph
445 big top burning the true story of an arsonist a missing girl
446 stay vigilant civil liberties are under constant attack nativehuman
myreligion
447 credit to for inspiring me to rediscover this fantabulous
tbt
448 nashville theater attack will gun grabbers now demand hatchet
control
449 breaking terror attack on police post udhampur
450 demi stans really think heart attack sold 5 6 million
copies
451 it scares me that there is new versions of nuclear attack warnings like just knowing
that governments stil...
452 isil claims suicide bombing at saudi mosque that killed at
least 15
453 the funniest part about that twitter is the feminists that try to attack
it 4head
454 horrific attack on wife by muslim in italy liveleak
news
455 93 blasts accused yeda yakub dies in karachi of heart attack
mumbai
456 us drone attack kills 4suspected militants in north
waziristan

457 suspect in latest theater attack had psychological
issues

458 militants attack police post in udhampur 2 spos injured livemint
allthenews

459 breaking obama officials gave muslim terrorist the weapon used in
texas attack

460 delhi government to provide free treatment to acid attack victims in
private hospitals

461 new post from new smb relay attack steals user credentials
over internet

462 israeli forces raid home of alleged car attack suspect
palestine

463 just had a heart attack because i thought my goat was dead do not
worry rocket is okay

464 i am not gonna lie i am kinda ready to attack my senior
year

465 left hand side of a diamond is a graveyard shift have to attack and defend the right
handside no have to ...

466 volleyball attack ii volleyball training machine sets
simulation

467 notley is tactful yet very direct response to harper is attack on alberta is gov t hell
yeah premier abl...

468 police assailant in latest us movie theatre attack was homeless had
psychological issues

469 thanks for the damn heart attack

470 suspect in latest us theatre attack had psychological
issues

471 india shud not give any evidence 2 pak they will share with terrorists and use for
next attack share wit...

472 illegal alien released by obama dhs 4 times charged with rape and murder of santa
maria ca woman had pri...

473 strongly condemn attack on ary news team in karachi a cowardly act against those
simply trying to do thei...

474 nashville theater attack will gun grabbers now demand hatchet
control

475 the fact that the atomic bombs were called little boy and fat man says a lot about the
mentality that went...

476 not everyone can see ignorance she is latino and that is all she can ever be nothing
more but an attack dog 4...

477 heart disease prevention what about secondhand
smoke

478 a daytona area org tells me it was hit by a cyber attack

479 attack on titan game on ps vita yay ca not wait for
2016

480 infowars nashville theater attack will gun grabbers now demand
hatchet control nwo

481 anxiety attack

482 my dog attacked me for my food pug probs

483 cop injured in gunfight as militants attack udhampur police post suspected militants
attacked a police pos...

484 i asked how did he feel attacked by julie i asked if he was frail that
is all

485 i feel attacked

486 i am feeling so attacked

487 once again black men did not make it that way white men did so why are black
men getting attacked

488 i cant believe a fucking cis female is going to somehow claim to be offended over a
transgendered female w...

489 israeli helicopters that attacked civilians in gaza just completed
exercises in greece

490 christian attacked by muslims at the temple mount after waving israeli flag via
pamela geller

491 christian attacked by muslims at the temple mount after waving israeli flag via
pamela geller

492 christian attacked by muslims at the temple mount after waving israeli flag via
pamela geller

493 pt the unit attacked by is was responsible for targeting muslim scholars and
imprisoning the youth

494 telnet attacked from 124 13 172 40
streamyx homesouthern my

495 christian attacked by muslims at the temple mount after waving israeli flag via
pamela geller

496 tbt remember that time patrick kane attacked a cab driver
over 20

497 im feeling attacked

498 ik only troll his pol rivals never literally abused them or attacked their families while
all of them lit...

499 christian attacked by muslims at the temple mount after waving israeli flag via
pamela geller

Name: text, dtype: object

In [18]:

```
from textblob import TextBlob
#Create a function to get the subjectivity
def getSubjectivity(twt):
    return TextBlob(twt).sentiment.subjectivity
```

In [19]:

```
df = train.text.apply(getSubjectivity)
```

In [20]:

```
import nltk
```

In [21]:

```
#nltk.download('punkt')
#nltk.download('wordnet')
#nltk.download('omw-1.4')
#nltk.download('stopwords')
```

In [22]:

```
train.text.head()
```

Out[22]:

0 our deeds are the reason of this earthquake may allah forgive
us all

1 forest fire near la ronge sask canada

2 all residents asked to shelter in place are being notified by officers no other
evacuation or shelter in ...

3 13000 people receive wildfires evacuation orders in
california

4 just got sent this photo from ruby alaska as smoke from wildfires pours
into a school

Name: text, dtype: object

Semantic Analysis - Removing Stop Words

In [23]:

```
from nltk.corpus import stopwords
```

In [24]:

```
from gensim.parsing.preprocessing import remove_stopwords
```

In [25]:

```
#stop_words = set(stopwords.words('english'))
"""def removeStopWords(tweet): filtered_sentence = [] i = 0 while i < len(tweet): for word
in tweet: if word not in stop_words: filtered_sentence.append(word) i = i + 1 return
filtered_sentence"""
```

In [26]:


```
#train.col_lemma = train.col_lemma.apply(lambda x: [item for item in x if item not in stop_words])
```

In [27]:

```
train
```

Out[27]:

	id	keyword	location	
0	1	NaN	NaN	our deeds are the reason of this earthquake may allah forgive us all
1	4	NaN	NaN	forest fire near la ronge sask canada
2	5	NaN	NaN	all residents asked to shelter in place are being notified by officers no other evacuation shelter in ...
3	6	NaN	NaN	13000 people receive wildfires evacuation orders in california
4	7	NaN	NaN	just got sent this photo from ruby alaska as smoke from wildfires pours into a school
...	...	...	...	...
7608	10869	NaN	NaN	two giant cranes holding a bridge collapse into nearby homes
7609	10870	NaN	NaN	ahrary the out of control wild fires in california even in the northern part of the state v troubling
7610	10871	NaN	NaN	m1 94 0104 utc 5km s of volcano hawaii
7611	10872	NaN	NaN	police investigating after an ebike collided with a car in little portugal ebike rider suffe serious n...
7612	10873	NaN	NaN	the latest more homes razed by northern california wildfire abc news

7613 rows × 5 columns

Tokenization and Lemmatization

In [28]:

```
test.text
```

Out[28]:

0	just happened a terrible car crash
1	heard about earthquake is different cities stay safe everyone

```

2           there is a forest fire at spot pond geese are fleeing across the street i
cannot save them all
3           apocalypse lighting  spokane wildfires
4           typhoon soudelor kills 28 in china and taiwan

...

3258           earthquake safety los angeles  safety fasteners
xrwn
3259  storm in ri worse than last hurricane  my city and 3others hardest hit  my yard
looks like it was bombed ...
3260           green line derailment in chicago
3261           meg issues hazardous weather outlook  hwo
3262           cityofcalgary has activated its municipal emergency plan
yycstorm
Name: text, Length: 3263, dtype: object

```

In [29]:

```

from nltk.tokenize import word_tokenize

train.text = train.text.apply(word_tokenize)
train.text.head()

test.text = test.text.apply(word_tokenize)
test.text.head()

```

Out[29]:

```

0           [our, deeds, are, the, reason, of, this, earthquake, may, allah, forgive,
us, all]
1           [forest, fire, near, la, ronge, sask, canada]
2  [all, residents, asked, to, shelter, in, place, are, being, notified, by, officers, no, other,
evacuation,...
3           [13000, people, receive, wildfires, evacuation, orders, in,
california]
4  [[just, got, sent, this, photo, from, ruby, alaska, as, smoke, from, wildfires, pours,
into, a, school]
Name: text, dtype: object

```

Out[29]:

```

0           [just, happened, a, terrible, car, crash]
1           [heard, about, earthquake, is, different, cities, stay, safe,
everyone]
2  [there, is, a, forest, fire, at, spot, pond, geese, are, fleeing, across, the, street, i, can,
not, save, ...
3           [apocalypse, lighting, spokane, wildfires]
4           [typhoon, soudelor, kills, 28, in, china, and, taiwan]

```

Name: text, dtype: object

In [30]:

```
from nltk.stem import WordNetLemmatizer

wnl = WordNetLemmatizer()

def lemmatize(s):
    s = [wnl.lemmatize(word) for word in s]
    return s

train = train.assign(text = train.text.apply(lambda x: lemmatize(x)))
train.text = train.text.squeeze()

test = test.assign(text = test.text.apply(lambda x: lemmatize(x)))
test.text = test.text.squeeze()
```

In [31]:

```
train.text.head(10)
```

Out[31]:

```
0          [our, deed, are, the, reason, of, this, earthquake, may, allah, forgive, u,
all]
1                                [forest, fire, near, la, ronge, sask, canada]
2  [all, resident, asked, to, shelter, in, place, are, being, notified, by, officer, no, other,
evacuation, o...
3                                [13000, people, receive, wildfire, evacuation, order, in,
california]
4  [just, got, sent, this, photo, from, ruby, alaska, a, smoke, from, wildfire, pours,
into, a, school]
5  [rockyfire, update, california, hwy, 20, closed, in, both, direction, due, to, lake, county,
fire, cafire,...
6  [flood, disaster, heavy, rain, cause, flash, flooding, of, street, in, manitou, colorado,
spring, area]
7                                [i, am, on, top, of, the, hill, and, i, can, see, a, fire, in, the, wood]
8  [there, is, an, emergency, evacuation, happening, now, in, the, building,
across, the, street]
9                                [i, am, afraid, that, the, tornado, is, coming, to, our, area]
```

Name: text, dtype: object

In [32]:

```
stop_words = stopwords.words('english')
stop_words.remove('not')

train.text = train.text.apply(lambda words: [word for word in words if word
not in stop_words])
```

```
test.text = test.text.apply(lambda words: [word for word in words if word not in stop_words])
```

In [33]:

```
train.head(5)
test.head(5)
```

Out[33]:

	i	keywor	locatio		text	target
	d	d	n			
0	1	NaN	NaN	[deed, reason, earthquake, may, allah, forgive, u]		1
1	4	NaN	NaN	[forest, fire, near, la, ronge, sask, canada]		1
2	5	NaN	NaN	[resident, asked, shelter, place, notified, officer, evacuation, shelter, place, order, expected]		1
3	6	NaN	NaN	[13000, people, receive, wildfire, evacuation, order, california]		1
4	7	NaN	NaN	[got, sent, photo, ruby, alaska, smoke, wildfire, pours, school]		1

Out[33]:

	id	keywor	locatio		text
		d	n		
0	0	NaN	NaN	[happened, terrible, car, crash]	
1	2	NaN	NaN	[heard, earthquake, different, city, stay, safe, everyone]	
2	3	NaN	NaN	[forest, fire, spot, pond, goose, fleeing, across, street, not, save]	
3	9	NaN	NaN	[apocalypse, lighting, spokane, wildfire]	
4	1 1	NaN	NaN	[typhoon, soudelor, kill, 28, china, taiwan]	

Full Cross-Validation Format Setup

In [34]:

```
import spacy, torch, transformers
from transformers import BertModel, BertTokenizer, TFBertModel,
TFBertForSequenceClassification, BertConfig, AdamW,
get_linear_schedule_with_warmup
```

In [35]:

```
train.text = train.text.apply(lambda x : " ".join(x))
```

In [36]:

```
from keras.preprocessing.text import Tokenizer
```

```
from keras.preprocessing.sequence import pad_sequences
```

```
tokenizer = Tokenizer(num_words=500, split=' ')\ntokenizer.fit_on_texts(train['text'].values)\nX_train = tokenizer.texts_to_sequences(train['text'].values)\nX_train = pad_sequences(X_train, maxlen=15)
```

In [37]:

```
X_test = tokenizer.texts_to_sequences(test['text'].values)\nX_test = pad_sequences(X_test, maxlen=15)
```

In [38]:

```
pd.DataFrame(X_train).head(5)
```

Out[38]:

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
0	0	0	0	0	0	0	0	0	0	0	0	46 2	16 5	67	6
1	0	0	0	0	0	0	0	0	0	0	0	0	11 2	3	15 3
2	0	0	0	0	0	0	0	0	0	0	46 3	33 2	16 6	46 3	33 3
3	0	0	0	0	0	0	0	0	0	0	12	79	16 6	33 3	33
4	0	0	0	0	0	0	0	0	0	0	31	11 0	19 0	79	10 4

In [39]:

```
pd.DataFrame(X_test).head(5)
```

Out[39]:

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
0	0	0	0	0	0	0	0	0	0	0	0	0	0	46	32
1	0	0	0	0	0	0	0	0	0	0	38 8	16 5	10 5	43 3	14 6
2	0	0	0	0	0	0	0	0	0	0	11 2	3	44 9	1	25 9
3	0	0	0	0	0	0	0	0	0	0	0	0	0	39 0	79

4 0 0 0 0 0 0 0 0 0 0 0 0 0 0 41 10 25
4 8 2

#1 LSTM Build.

In [40]:

```
from keras.layers import Dense, Embedding, LSTM, SpatialDropout1D

model = Sequential()
model.add(Embedding(500, 120, input_length = X_train.shape[1]))
model.add(SpatialDropout1D(0.4))
model.add(LSTM(176, dropout=0.2, recurrent_dropout=0.2))
model.add(Dense(1,activation='sigmoid'))
model.compile(loss = 'binary_crossentropy', optimizer='adam', metrics =
['accuracy'])
print(model.summary())
Model: "sequential"
```

Layer (type)	Output Shape	Param #
=====		
embedding (Embedding)	(None, 15, 120)	60000
spatial_dropout1d (SpatialD ropout1D)	(None, 15, 120)	0
lstm (LSTM)	(None, 176)	209088
dense (Dense)	(None, 1)	177
=====		
Total params: 269,265		
Trainable params: 269,265		
Non-trainable params: 0		
None		

2022-05-29 23:16:48.589339: I tensorflow/core/platform/cpu_feature_guard.cc:151] This TensorFlow binary is optimized with oneAPI Deep Neural Network Library (oneDNN) to use the following CPU instructions in performance-critical operations: SSE4.1 SSE4.2 To enable them in other operations, rebuild TensorFlow with the appropriate compiler flags.

In [41]:

```
from sklearn.model_selection import train_test_split

train_valid, test_valid = train_test_split(X_train, random_state=42,
train_size=0.8)
print(train_valid.shape)
print(test_valid.shape)
(6090, 15)
(1523, 15)
```

In [42]:

```
y_train_valid, y_test_valid = train_test_split(train.target, random_state=42,
train_size=0.8)
```

In [43]:

```
train_valid[0:5]
test_valid[0:5]
y_train_valid.head(5)
y_test_valid.head(5)
```

Out[43]:

```
array([[ 0,  0,  0,  0,  0,  0,  0,  0,  0,  0,  0, 64, 402, 144,
        36, 177],
       [ 0,  0,  0,  0,  0,  0,  0,  0,  0,  0,  0,  0,  0, 368,
        380, 145],
       [ 0,  0,  0,  0,  0,  0,  0,  0,  0,  0,  0, 491,  6, 382,
        203, 291],
       [ 0,  0,  0,  0,  0,  0,  0,  0,  0,  0,  0,  0, 341, 308,
        175, 111],
       [ 0,  0,  0,  0,  0,  0,  0,  0,  0,  0,  0,  0, 420, 407,
        371, 338]], dtype=int32)
```

Out[43]:

```
array([[ 0,  0,  0,  0,  0,  0,  0,  0,  0,  0,  0,  0,  8, 72,
        130, 303],
       [ 0,  0,  0,  0,  0,  0,  0,  0,  0,  0,  0,  0, 92, 31,
        349, 53],
       [ 0,  0,  0,  0,  0,  0,  0,  0,  0,  0,  0,  0,  0,  0,
         0, 22],
       [ 0,  0,  0,  0,  0, 34, 104,  2, 127, 62, 425, 146, 450,
        127, 172],
       [ 0,  0,  0,  0,  0,  0,  0,  0,  0,  0,  0,  0, 302, 201,
        10, 47]], dtype=int32)
```

Out[43]:

```
4996  1
3263  0
```

4907 1
2855 1
4716 0

Name: target, dtype: int64

Out[43]:

2644 1
2227 0
5448 1
132 0
6845 0

Name: target, dtype: int64

In [44]:

```
batch_size=10  
model.fit(train_valid, y_train_valid, epochs = 10, batch_size=batch_size,  
validation_data =(test_valid, y_test_valid) , verbose = 'auto')
```

Epoch 1/10

609/609 [=====] - 7s 10ms/step - loss: 0.5353 - accuracy:
0.7263 - val_loss: 0.5016 - val_accuracy: 0.7643

Epoch 2/10

609/609 [=====] - 6s 10ms/step - loss: 0.4531 - accuracy:
0.7943 - val_loss: 0.5032 - val_accuracy: 0.7571

Epoch 3/10

609/609 [=====] - 6s 10ms/step - loss: 0.4340 - accuracy:
0.8028 - val_loss: 0.5135 - val_accuracy: 0.7656

Epoch 4/10

609/609 [=====] - 6s 10ms/step - loss: 0.4249 - accuracy:
0.8087 - val_loss: 0.5214 - val_accuracy: 0.7544

Epoch 5/10

609/609 [=====] - 7s 11ms/step - loss: 0.4133 - accuracy:
0.8122 - val_loss: 0.5233 - val_accuracy: 0.7623

Epoch 6/10

609/609 [=====] - 6s 11ms/step - loss: 0.3953 - accuracy:
0.8200 - val_loss: 0.5306 - val_accuracy: 0.7525

Epoch 7/10

609/609 [=====] - 7s 11ms/step - loss: 0.3813 - accuracy:
0.8241 - val_loss: 0.5383 - val_accuracy: 0.7590

Epoch 8/10

609/609 [=====] - 7s 11ms/step - loss: 0.3652 - accuracy:
0.8342 - val_loss: 0.5393 - val_accuracy: 0.7663

Epoch 9/10

609/609 [=====] - 7s 11ms/step - loss: 0.3485 - accuracy:
0.8415 - val_loss: 0.5716 - val_accuracy: 0.7518
Epoch 10/10
609/609 [=====] - 7s 11ms/step - loss: 0.3345 - accuracy:
0.8501 - val_loss: 0.5896 - val_accuracy: 0.7406

Out[44]:

<keras.callbacks.History at 0x1ccfb2f20>

In [45]:

```
y_pred = model.predict(X_test)

threshold =0.5
y_pred = np.where(y_pred > threshold, 1, 0)
y_pred = pd.DataFrame(y_pred)
y_pred.columns = ['target']
```

In [46]:

```
y_pred.head(100)
```

Out[46]:

	target
0	1
1	1
2	1
3	1
4	1
5	1
6	0
7	0
8	0
9	0
10	0

1
1 0

1
2 0

1
3 0

1
4 0

1
5 0

1
6 0

1
7 0

1
8 0

1
9 0

2
0 0

2
1 1

2
2 0

2
3 0

2
4 0

2
5 1

2
6 0

2
7 0

2
8 0

2
9 1

3
0 0

3
1 0

3
2 1

3
3 1

3
4 1

3
5 0

3
6 1

3
7 0

3
8 0

3
9 1

4
0 1

4
1 1

4
2 0

4
3 0

4
4 0

4
5 0

4
6 1

4
7 0

4
8 0

4
9 0

5
0 0

5
1 0

5
2 1

5
3 0

5
4 0

5
5 0

5
6 0

5
7 0

5
8 0

5
9 0

6
0 1

6
1 1

6
2 1

6
3 1

6
4 1

6
5 1

6
6 1

6
7 1

6
8 1

6
9 1

7
0 1

7
1 0

7
2 1

7
3 1

7
4 1

7 1
5

7 0
6

7 0
7

7 1
8

7 1
9

8 1
0

8 0
1

8 0
2

8 0
3

8 1
4

8 0
5

8 0
6

8 0
7

8 1
8

8 1
9

9 0
0

9 0
1

9 0
2

9 0
3

9 1
4

9 0
5

9 0
6

9 0
7

9 0
8

9 0
9

In [47]:

```
submission = pd.DataFrame(test.id)
```

In [48]:

```
submission['target'] = y_pred
```

In [49]:

```
submission = submission.set_index('id')
```

In [50]:

```
submission.target.value_counts()
```

Out[50]:

0 2036

1 1227

Name: target, dtype: int64

In [51]:

```
submission.head(10)
```

Out[51]:

target
t

id

0 1

2 1

3 1

9 1

1 1
1 1

1 1
2 1

2 0
1 0

2 0
2 0

2 0
7 0

2 0
9 0

In [52]:

```
submission.to_csv("submission.csv")
```

First Submission Kaggle Score: 0.75543

#2 LSTM Build.

In [53]:

```
max_features = 3000
embed_dim = 32
lstm_out = 32
model = Sequential()
model.add(Embedding(max_features, embed_dim, input_length = X_train.shape[1]))
model.add(Dropout(0.2))
model.add(LSTM(lstm_out, dropout=0.2, recurrent_dropout=0.4))
model.add(Dense(1, activation='sigmoid'))
adam = tensorflow.keras.optimizers.Adam(learning_rate=0.002)
model.compile(loss = 'binary_crossentropy', optimizer=adam, metrics =
['accuracy'])
print(model.summary())
Model: "sequential_1"
```

Layer (type)	Output Shape	Param #
embedding_1 (Embedding)	(None, 15, 32)	96000
dropout (Dropout)	(None, 15, 32)	0
lstm_1 (LSTM)	(None, 32)	8320
dense_1 (Dense)	(None, 1)	33

Total params: 104,353
 Trainable params: 104,353
 Non-trainable params: 0

None

In [54]:

```

batch_size=64
model.fit(train_valid, y_train_valid, epochs = 5, batch_size=batch_size,
validation_data =(test_valid, y_test_valid) , verbose = 'auto')
Epoch 1/5
96/96 [=====] - 2s 11ms/step - loss: 0.5948 - accuracy:
0.6778 - val_loss: 0.4962 - val_accuracy: 0.7663
Epoch 2/5
96/96 [=====] - 1s 10ms/step - loss: 0.4557 - accuracy:
0.7911 - val_loss: 0.4967 - val_accuracy: 0.7689
Epoch 3/5
96/96 [=====] - 1s 10ms/step - loss: 0.4356 - accuracy:
0.8000 - val_loss: 0.5034 - val_accuracy: 0.7682
Epoch 4/5
96/96 [=====] - 1s 10ms/step - loss: 0.4284 - accuracy:
0.8100 - val_loss: 0.5058 - val_accuracy: 0.7617
Epoch 5/5
96/96 [=====] - 1s 10ms/step - loss: 0.4237 - accuracy:
0.8112 - val_loss: 0.5139 - val_accuracy: 0.7669

```

Out[54]:

<keras.callbacks.History at 0x1ced5a830>

In [55]:

```

y_pred = model.predict(X_test)

threshold =0.5
y_pred = np.where(y_pred > threshold, 1, 0)
y_pred = pd.DataFrame(y_pred)
y_pred.columns = ['target']
submission = pd.DataFrame(test.id)
submission['target'] = y_pred
submission = submission.set_index('id')
submission.target.value_counts()

```

Out[55]:

```

0    2125
1    1138
Name: target, dtype: int64

```

In [56]:

```

submission.to_csv("submission.csv")

```

Second Submission Kaggle Score: 0.77290

#3 LSTM Build

In [69]:

```

max_features = 800
embed_dim = 64
lstm_out = 32
model = Sequential()
model.add(Embedding(max_features, embed_dim,input_length = X_train.shape[1]))
model.add(Dropout(0.2))
model.add(LSTM(lstm_out, dropout=0.2, recurrent_dropout=0.4))
model.add(Dense(1,activation='sigmoid'))
adam = tensorflow.keras.optimizers.Adam(learning_rate=0.002)
model.compile(loss = 'binary_crossentropy', optimizer=adam ,metrics =
['accuracy'])
print(model.summary())
Model: "sequential_5"

```

Layer (type)	Output Shape	Param #
=====		
embedding_5 (Embedding)	(None, 15, 64)	51200
dropout_4 (Dropout)	(None, 15, 64)	0
lstm_5 (LSTM)	(None, 32)	12416
dense_5 (Dense)	(None, 1)	33

=====

Total params: 63,649

Trainable params: 63,649

Non-trainable params: 0

None

In [70]:

```
batch_size=16
model.fit(train_valid, y_train_valid, epochs = 3, batch_size=batch_size,
validation_data =(test_valid, y_test_valid) , verbose = 'auto')
```

Epoch 1/3

381/381 [=====] - 5s 9ms/step - loss: 0.5311 - accuracy:
0.7338 - val_loss: 0.5002 - val_accuracy: 0.7538

Epoch 2/3

381/381 [=====] - 3s 9ms/step - loss: 0.4504 - accuracy:
0.7903 - val_loss: 0.5105 - val_accuracy: 0.7623

Epoch 3/3

381/381 [=====] - 3s 9ms/step - loss: 0.4348 - accuracy:
0.8005 - val_loss: 0.5058 - val_accuracy: 0.7564

Out[70]:

<keras.callbacks.History at 0x1d4655960>

In [71]:

```
y_pred = model.predict(X_test)

threshold =0.5
y_pred = np.where(y_pred > threshold, 1, 0)
y_pred = pd.DataFrame(y_pred)
y_pred.columns = ['target']
submission = pd.DataFrame(test.id)
submission['target'] = y_pred
submission = submission.set_index('id')
submission.target.value_counts()
```

Out[71]:

0 2205

1 1058

Name: target, dtype: int64

In [72]:

```
submission.to_csv("submission.csv")
```

Third Submission Kaggle Score: 0.77137

